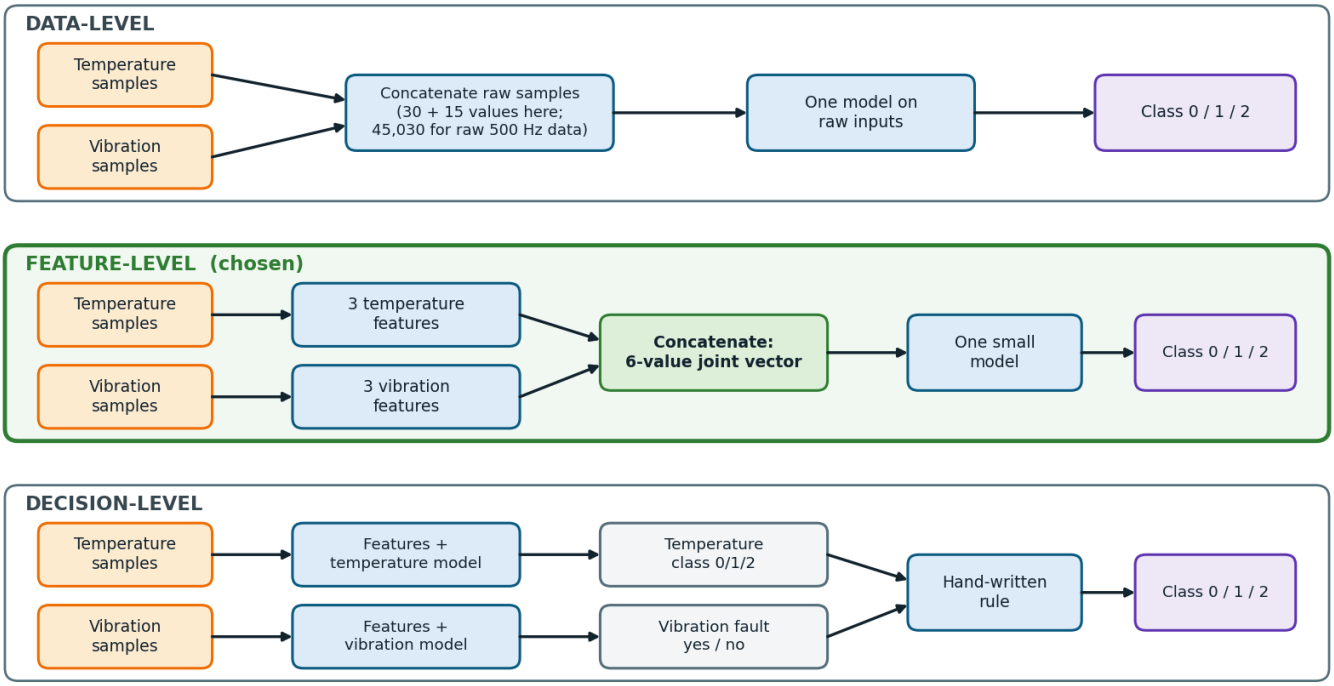
Task C3 — Data Fusion Justification

Final Report section | LogiEdge | FreightBridge Logistics Pvt. Ltd. | Component C, Module 3

At a glance

Question	Answer
Fusion level chosen	**Feature-level.** Each sensor is summarised on its own into 3 features; the two blocks are concatenated into one 6-value joint vector for one small model.
Against data-level	Same accuracy (99.7% against 99.6%) at about one fifth of the compute (144 against 768 multiply-accumulates). On raw 500 Hz vibration the gap grows to about 5,003 times.
Against decision-level	A simple max rule loses 2.0 points (97.7%); matching accuracy (99.7%) needs a hand-written cross-sensor rule that feature-level learns from data.
Honest limit	Accuracy differences between sound designs are within 0.25 points in simulation. The case rests on cost, simplicity and fit, not on a large accuracy gain.

1 The three fusion levels



Level	Where the sensors are combined	Typical use
Data-level	Raw samples are joined before any feature or model	Same kind of sensor, same rate, well aligned
Feature-level	Each sensor is reduced to features; the features are joined	Different sensors with different rates and units
Decision-level	Each sensor reaches its own decision; the decisions are combined	Independent sensors, or when modules must stay separate

2 What was implemented

```
t = temperature_features(ts, temp_window)   # [mean, std, rate in degC/min]   (temperature only)
v = vibration_features(vib_window)          # [RMS, peak, kurtosis]            (vibration only)
joint = fuse_features(t, v)                 # one 6-value vector: t first, then v
z = normaliser.transform(joint)             # saved statistics from training_stats.npy
```

- Features are extracted independently: a test changes only the vibration stream and confirms the three temperature features do not move, and the reverse.
- Concatenation happens before the model, so the model sees one 6-value input and can learn how the two sensors interact.

3 Justification for this application

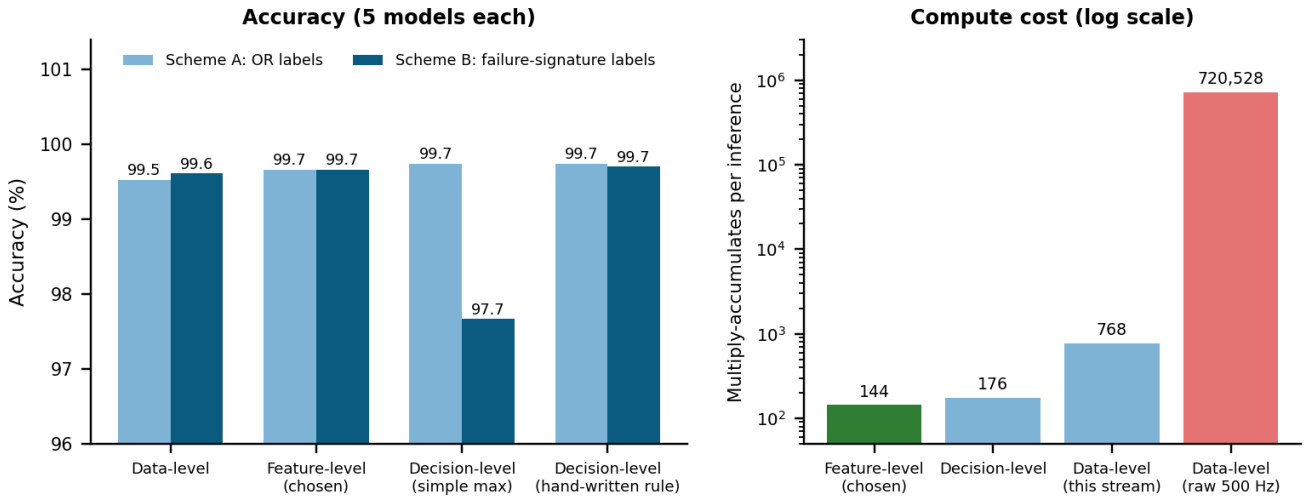
Criterion	Data-level	Feature-level (chosen)	Decision-level
Different sensors and rates	Needs alignment of 1 Hz °C readings, 0.5 Hz g values and, in the real truck, 500 Hz × 3 axes, plus unit scaling	Each stream is summarised over the same 30 s window every 10 s, so the rates never meet	No alignment problem
Edge compute (10 W budget, Tasks B1 and B2)	768 multiply-accumulates here; about 720,528 on raw 500 Hz data (45,030 inputs)	144 multiply-accumulates; 6 inputs	176 multiply-accumulates across two models
Cross-sensor meaning (a failing compressor with rising temperature is a unit-failure signature)	Can learn it, from many more inputs	Learns it directly from the joint vector	Needs a hand-written rule; the simple max rule loses 2.0 points
Audit and explanation (pharma chain of custody)	Opaque inputs	Each input is physical: mean °C, rate °C/min, RMS g. Vectors can be logged with alerts	Per-sensor decisions are easy to explain
Fit with Task C2 statistics	45 or 45,030 statistics to store and keep consistent	Six, already in training_stats.npy	Six, split across two models
Sensor failure	One input group disappears	Window is skipped and counted; needs a watchdog (Section 5)	Easy to fall back to the other sensor in principle
Information kept	Everything, including waveform shape	Summary only; add features if needed	Summary only, then reduced to a class

4 Evidence from the simulator

Same data, labels and training budget for every design (16 hidden units, 5 models each). **Scheme A** labels a window by the worse of the two sensors (an "OR"). **Scheme B** also labels a vibration fault together with temperature more than 1 °C off setpoint as Class 2, a unit-failure signature. Test set: 3,240 windows.

Design	Acc. A	Acc. B	Params	MACs	No vib. (B)	No temp. (B)
Data-level (30 + 15 samples)	99.53%	99.62%	787	768	77.0%	64.2%
Feature-level (chosen)	99.67%	99.67%	163	144	77.2%	64.1%
Decision-level, simple max	99.75%	97.67%	213	176	77.2%	64.2%
Decision-level, hand-written rule	99.75%	99.71%	213	176	77.2%	64.2%

Acc. = accuracy on that labelling scheme; MACs = multiply-accumulates per inference; No vib. / No temp. = Scheme B accuracy when that sensor's inputs are replaced by Normal values.



The accuracy axis starts at 96% so small differences are visible. The raw 500 Hz bar is calculated (45,030 inputs × 16 hidden units + 16 × 3), not trained.

What the numbers show

- **Accuracy is essentially tied.** Data-level, feature-level and rule-based decision-level are all within 0.25 points on both schemes. I am not claiming feature-level is more accurate.
- **Compute separates them.** Feature-level needs 144 multiply-accumulates against 768 for data-level on this stream, and about 720,528 on raw 500 Hz vibration, roughly 5,003 times more.
- **Decision-level needs the cross-sensor logic written by hand.** With simple max it drops from 99.75% on Scheme A to 97.67% on Scheme B, because it cannot see that the two anomalies together mean failure. A correct rule recovers the accuracy but has to be rewritten and revalidated whenever the class definitions change; feature-level only needs retraining.
- **Missing-sensor tests found no difference.** When one sensor's inputs are replaced by Normal values, all four designs lose the same accuracy (within 0.3 points), since the missing sensor's anomalies are simply unseen. Decision-level's fault-tolerance advantage did not appear in this test.

5 Where feature-level is weaker, and how it is covered

Weakness	Cover
Waveform detail is discarded	The current vibration stream is already RMS. If bearing diagnosis later needs frequency content, add spectral features to the vector before concatenation; the approach extends without redesign.
A failed sensor can pass unnoticed	The pipeline skips and counts windows with under 80% of their samples. Add a watchdog that raises a sensor-fault alert when windows are skipped, rather than classifying on missing data.
Hand-picked features may miss a new failure mode	Keep fleet-level drift and accuracy monitoring (the MLOps component) and retrain when new faults appear.
Sensitive to normalisation statistics	Task C2 showed a 3σ shift costs 1.3 to 8.4 points, so the statistics are fixed, loaded from file and never recomputed from live data.

6 Conclusion

Feature-level fusion is the best fit for LogiEdge. The two sensors differ in rate, unit and meaning, the truck has a 10 W compute budget, and, under the failure-signature reading of Class 2, the class depends on how temperature and vibration combine. Summarising each sensor first keeps the model tiny and its inputs physical and loggable for pharmaceutical audit, while the joint vector lets one model learn the cross-sensor logic without hand-written rules. Data-level fusion would match the accuracy at far higher cost and decision-level fusion at the price of maintained rules; neither offers a measured accuracy benefit here.

7 Limits and items to confirm

- Results come from simulated data with a small reference classifier. Rerun `experiments_c2_c3.py` with your trained model and real truck data before treating the accuracy comparison as final.
- Scheme B is my interpretation of “refrigeration unit failure signature”. If your labels follow a pure OR rule (Scheme A), decision-level with a max rule is as accurate and the argument rests on compute and maintenance only.
- The data-level comparison uses 30 + 15 filtered samples. The raw 500 Hz cost is a calculation with 16 hidden units, not a trained model.
- All designs use 16 hidden units; other sizes would change the compute figures but not their ordering.